Enhancing U-Net for Visual Saliency Prediction with Multi-Scale Attention

Alessio Zattoni[†]

***Abstract*—Visual saliency prediction aims to model human visual attention by identifying the regions in an image that are most likely to attract gaze. It is a useful task in computer vision with applications in image understanding, compression, and human-computer interaction. Although deep learning methods have significantly improved performance in recent years, accurately capturing both local details and global context in complex scenes remains challenging.**

This work is based on a standard U-Net architecture and investigates simple modifications to improve its ability to model multi-scale information. Inception modules are introduced in the bottleneck to extract features at different receptive fields, allowing the network to better handle objects of varying sizes. In addition, Squeeze-and-Excitation blocks are used to recalibrate channel-wise feature responses, helping the model focus on more relevant information. A lightweight fusion mechanism is also added to combine features before decoding.

The main goal is not to significantly increase model complexity, but to evaluate whether relatively small architectural changes can improve performance in a standard encoder-decoder framework. Experiments are conducted on the SALICON dataset, a widely used benchmark for saliency prediction. Results show that the proposed model consistently improves over a baseline U-Net, producing more accurate and spatially focused saliency maps.

Overall, the study suggests that combining multi-scale feature extraction with lightweight attention mechanisms can effectively enhance saliency prediction without introducing a high computational cost.

***Index Terms*—Visual Saliency, U-Net, skip connections, Deep Learning, Inception Network, Squeeze-and-Excitation**

I. INTRODUCTION

Visual saliency prediction is a fundamental problem in computer vision that aims to model human visual attention by identifying regions in an image that are most likely to attract visual focus. It plays a key role in a wide range of applications, including image understanding, compression, visual tracking and human-computer interaction. In recent years, deep learning methods have significantly improved the ability to learn saliency patterns directly from data, making this problem increasingly relevant in modern vision systems.

Despite recent progress in deep learning-based saliency prediction, accurately modeling human attention in complex scenes remains challenging. Encoder-decoder architectures such as U-Net [1], often enhanced with skip connections, have become strong baselines due to their ability to combine low-level spatial details with high-level semantic features. However, standard U-Net-based approaches may still struggle to effectively capture multi-scale contextual information and long-range dependencies, which are essential in scenes containing objects of varying sizes and cluttered backgrounds.

To overcome these limitations, an improved U-Net-based architecture is proposed, extending the standard encoder-decoder framework with enhanced feature representation. The method integrates Inception modules [2] into the bottleneck to model multi-scale context, Squeeze-and-Excitation (SE) blocks [3] to perform adaptive channel-wise feature recalibration, and a self-excitation fusion mechanism to increase feature diversity by combining representations with different receptive fields at the same semantic level. Compared to standard U-Net-based saliency models [1], the proposed design improves the ability to capture both local and global information while maintaining a compact structure.

The main advantage of the proposed approach lies in improving representational capacity without significantly increasing computational complexity, addressing a key limitation of existing saliency methods that rely either on shallow multi-scale processing or computationally expensive attention mechanisms.

The proposed architecture can be applied to large-scale saliency prediction tasks and integrated into downstream vision systems requiring reliable visual attention estimation, such as scene understanding, image compression, and intelligent visual analytics.

The main contributions of this work are summarized as follows:

- An improved U-Net-based architecture for visual saliency prediction is proposed, enhancing feature representation within an encoder-decoder framework.
- Inception modules are integrated into the bottleneck to enable effective multi-scale contextual feature extraction.
- Squeeze-and-Excitation (SE) blocks are introduced to perform adaptive channel-wise feature recalibration.
- A self-excitation fusion mechanism is designed to increase feature diversity by combining representations with different receptive fields at the same semantic level.

The paper is organized as follows: Section II reviews related work; Section III presents the proposed architecture and processing pipeline; Section IV describes the dataset and feature representation; Section V outlines the learning framework and training procedure; Section VI reports the experimental results; Section VII concludes the paper; and Section VIII discusses the main takeaways and challenges.

[†]Department of Information Engineering, University of Padova,
email: `alessio.zattoni@unipd.it`
ID: 2125280

II. Related Work

With the rise of deep learning, convolutional neural networks (CNNs) have significantly improved saliency prediction by learning hierarchical features directly from data. Fully convolutional networks and encoder-decoder architectures quickly became standard choices, mainly because they can produce dense pixel-wise predictions while still preserving the spatial structure of the image. Among them, U-Net [1] is widely used as a strong baseline thanks to its symmetric encoder-decoder design and skip connections, which help combine fine spatial details from early layers with higher-level semantic information. This makes it particularly well suited for dense prediction tasks such as segmentation and saliency estimation. Despite these strengths, CNN-based methods still have difficulty capturing global context and long-range dependencies, especially in complex scenes with multiple objects, cluttered backgrounds, and strong scale variations.

To overcome these issues, transformer-based models have recently become the leading approach in visual saliency prediction. By relying on self-attention, they can directly model interactions between all image regions, which allows them to better capture long-range dependencies compared to purely convolutional architectures. Vision Transformer (ViT)-based models [4] and their variants have achieved strong results across many vision tasks, including saliency prediction, largely because they can reason over the entire image in a more global way. This global modeling tends to align better with human visual attention, particularly in situations where saliency depends on contextual relationships spread across different parts of the scene. However, these advantages come at a cost, including higher computational requirements, increased memory usage, and the need for large-scale datasets during training. Because of this, transformer-based approaches are often less practical for real-time or resource-constrained scenarios.

In contrast, the proposed method tries to strike a balance between the efficiency of CNNs and some of the representational advantages of transformers. It is built on a U-Net-based architecture and incorporates multi-scale feature extraction along with attention mechanisms to improve contextual understanding without introducing excessive computational overhead. Specifically, it combines Inception-style multi-branch feature extraction, Squeeze-and-Excitation channel attention, and a dedicated feature fusion module to strengthen the learned representations. Overall, this design improves saliency prediction performance while keeping the model lightweight, achieving a better trade-off between accuracy and efficiency, and making it suitable for large-scale as well as potentially real-time applications.

III. Processing Pipeline

The proposed architecture builds on a supervised learning framework for visual saliency prediction and integrates multi-scale processing within an encoder-decoder structure. As shown in Fig. 1, the pipeline is organized into three main stages: feature extraction, bottleneck refinement and map reconstruction.

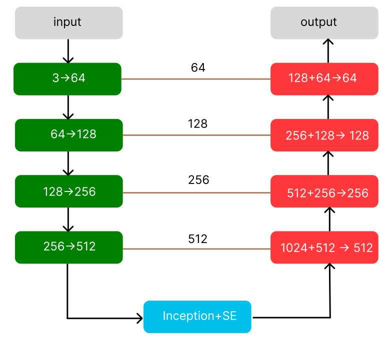

Fig. 1: Overall processing pipeline of the proposed Inception-SE U-Net for visual saliency prediction. Encoder blocks are shown in green, decoder blocks in red and the Inception-SE bottleneck block in cyan; skip connections are depicted as brown lines linking corresponding encoder and decoder stages.

A. Architectural Overview

The core of the system is the Inception-SE U-Net, an extension of the classical U-Net adapted for saliency prediction tasks [1]. The input is a normalized RGB image, which is then processed through the following components:

- **Encoder Stage:** A series of convolutional blocks and pooling layers gradually reduce spatial resolution while increasing feature dimensionality, allowing the model to build hierarchical representations of the input.
- **Multi-scale Bottleneck:** An Inception-based module [2] processes features in parallel branches with different receptive fields. The outputs are concatenated and then fused using a 1×1 convolution.
- **Squeeze-and-Excitation (SE) Block:** A channel-wise attention mechanism is applied within the bottleneck to recalibrate feature responses and highlight the most relevant channels [3].
- **Decoder Stage:** Upsampling layers combined with skip connections gradually merge high-level semantic information with spatial details preserved by the encoder, producing a full-resolution saliency map.

B. Interaction between Blocks

The encoder progressively compresses the input while preserving hierarchical feature representations. Skip connections help retain spatial information, which is further refined in the bottleneck (Fig. 2) and then passed to the decoder for reconstruction. The final output is a pixel-wise probability map obtained using a Sigmoid activation, representing the predicted fixation density.

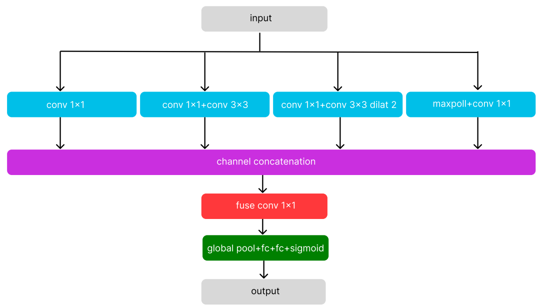

Fig. 2: Detailed architecture of the Inception-SE bottleneck module. Parallel convolutional branches are fused and refined using channel-wise attention. The four parallel branches of the Inception block are shown in cyan, the concatenation block in purple, the fusion block in red and the SE block in green.

IV. Signals and Features

This section describes the dataset characteristics, the preprocessing pipeline and the feature representation used for the saliency prediction task.

A. Measurement Setup and Data Formatting

The study is based on the SALICON dataset, a large-scale benchmark for visual saliency [5]. The input consists of natural images paired with corresponding human fixation maps, , available at https://www.kaggle.com/datasets/roshan401/salicon. The raw data is organized as follows:

- **Input Images:** RGB images.
- **Saliency Maps:** Single-channel grayscale images where pixel intensity indicates the degree of human fixation, serving as ground truth.

All image-map pairs are resized to a fixed resolution of 224×224 pixels to ensure consistent input dimensions across the network. This resolution is a commonly adopted standard in CNN-based vision models.

B. Signal Pre-processing

A structured preprocessing pipeline is applied before feeding the data into the model:

- **Resizing:** Both images and saliency maps are resized to $(224, 224)$.
- **Normalization:** RGB images are converted to tensors with pixel values scaled to $[0, 1]$. Saliency maps are converted to grayscale and normalized in the same range to represent a fixation probability distribution.
- **Data Augmentation:** During training, horizontal flipping is applied with probability 0.5, which helps reduce spatial bias and improves generalization.

C. Feature Extraction and Representation

Unlike time-series data, which typically requires temporal windowing, visual inputs are processed as spatial tensors. Feature representations are learned through an encoder-decoder architecture:

- **Local Features:** Early layers extract low-level patterns such as edges and textures using 3×3 convolutions with Batch Normalization and ReLU activation.
- **Global Context:** The bottleneck is enhanced using an Inception-style module [2], which captures multi-scale context through parallel branches with different receptive fields.
- **Feature Integration:** A Squeeze-and-Excitation (SE) block is used for channel-wise recalibration [3], allowing the model to emphasize the most informative features before decoding.

D. Dataset Splitting

The dataset is divided into three subsets to support training and evaluation in a consistent way. The split is defined as follows:

- **Training Set:** 10,000 image-map pairs used for model optimization.
- **Validation Set:** 2,500 pairs used for monitoring during training and early stopping.
- **Test Set:** 2,500 pairs reserved for final evaluation on unseen data.

This setup ensures a strict separation between training, validation and test sets.

V. Learning Framework

This section describes the learning strategy and optimization procedure used to train the proposed saliency prediction model. The framework follows a supervised learning approach aimed at minimizing the pixel-wise difference between predicted and ground-truth saliency maps [5].

A. Loss Function

The model is trained using the Mean Squared Error (MSE) loss, which quantifies the pixel-wise discrepancy between predicted and ground-truth saliency maps. Given a ground truth map $\boldsymbol{Y}$ and a prediction $\hat{\boldsymbol{Y}}$, the loss is defined as:

$$\mathcal{L}(\boldsymbol{Y}, \hat{\boldsymbol{Y}}) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2, \tag{1}$$

where N is the total number of pixels and y_i and $\hat{y}_i$ denote the ground truth and predicted pixel values, respectively. This objective encourages accurate pixel-level reconstruction of the saliency maps.

B. Optimization Strategy

Model optimization is carried out using the Adam optimizer, an adaptive gradient-based method. The training setup is defined as follows:

- **Learning Rate:** 10^{-4}, chosen to ensure stable convergence during training.
- **Batch Size:** 16, providing a trade-off between computational efficiency and gradient stability.
- **Weight Initialization:** Xavier (Glorot) initialization [6] is applied to convolutional layers to maintain stable variance across layers, while biases are initialized to zero.

C. Training Procedure and Regularization

The model is trained for up to 12 epochs, with validation performance monitored at the end of each epoch. To improve generalization and reduce overfitting, the following strategies are used:

- **Early Stopping:** Training is stopped if the validation loss does not improve for a fixed number of epochs (patience-based criterion [7]).
- **Model Checkpointing:** The model is saved at each epoch and the final evaluation is performed using the weights corresponding to the best validation performance.

D. Evaluation Metric

In addition to the MSE loss, Intersection over Union (IoU) is used as a secondary evaluation metric to measure spatial overlap between predicted and ground-truth saliency regions.

Predictions are binarized using a threshold of 0.5 applied after the sigmoid activation:

$$\hat{Y}_{bin} = \mathbb{I}(\sigma(\hat{Y}) > 0.5), \tag{2}$$

where $\sigma(\cdot)$ is the sigmoid function and $\mathbb{I}$ is the indicator function.

The IoU is then computed as:

$$IoU = \frac{|\hat{Y}_{bin} \cap Y|}{|\hat{Y}_{bin} \cup Y|}. \tag{3}$$

VI. Results

This section presents the experimental results of the proposed saliency prediction models, evaluated both quantitatively and qualitatively.

A. Learning Curves

Figure 3 shows the training and validation loss curves for both the baseline UNet and the proposed Inception-SE UNet. The proposed architecture achieves lower validation loss while maintaining a small gap between training and validation curves, indicating better generalization and stable convergence.

Fig. 3: Training and validation loss curves for the baseline UNet and the proposed Inception-SE UNet.

B. Quantitative Results

The performance of the proposed architecture is evaluated using Mean Squared Error (MSE) and Intersection over Union (IoU). MSE measures the pixel-wise discrepancy between predicted and ground-truth saliency maps, while IoU evaluates the spatial overlap between predicted and ground-truth salient regions after thresholding.

TABLE 1: Performance comparison on the test set. Relative improvements are computed with respect to the UNet baseline.

Model	Loss	IoU	Loss (%)	IoU (%)
UNet	0.0108	0.1371	–	–
Inc-SE UNet	0.0100	0.1638	-7.4	+19.5

As reported in Table 1, the proposed Inception-SE UNet consistently outperforms the baseline UNet. The integration of the Inception bottleneck and the Squeeze-and-Excitation module reduces the test loss from 0.0108 to 0.0100, corresponding to a 7.4% reduction, while the IoU increases from 0.1371 to 0.1638, yielding a 19.5% gain.

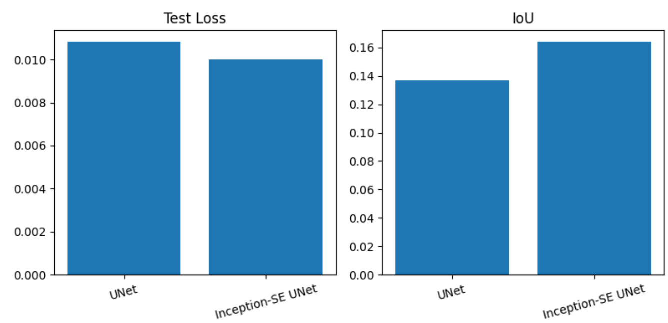

Fig. 4: Histogram of loss and IoU values for the baseline UNet on the test set, showing the distribution of evaluation metrics across samples.

The histogram 4 illustrates the distribution of evaluation metrics (loss and IoU) for the baseline UNet over the test set. It highlights the variability of prediction quality across samples, with most predictions concentrated around the reported average performance.

C. Qualitative Results

Figure-based evaluation highlights the ability of the proposed model to generate saliency maps that closely match human fixation patterns. In particular, the model consistently identifies high-attention regions that are better aligned with the ground truth annotations compared to the baseline UNet.

The proposed Inception-SE UNet produces more focused and spatially coherent saliency maps 6, reducing the presence of spurious or secondary activation regions that are often observed in the baseline predictions 5 . In contrast, the standard UNet tends to exhibit multiple activation modes, spreading attention across irrelevant areas of the scene.

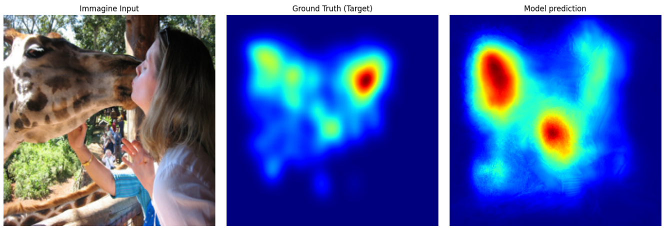

Fig. 5: Qualitative results of the baseline UNet. From left to right: input image, ground truth saliency map, and predicted saliency map.

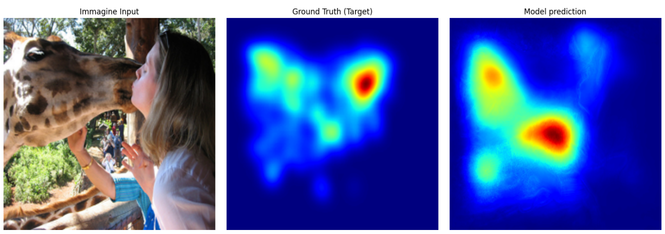

Fig. 6: Qualitative results of the proposed Inception-SE UNet. From left to right: input image, ground truth saliency map, and predicted saliency map.

VII. Concluding Remarks

This work presented a supervised deep learning framework for visual saliency prediction based on an encoder-decoder architecture. An Inception-SE U-Net was introduced to enhance multi-scale feature extraction and improve channel-wise feature recalibration compared to a standard U-Net baseline.

The experimental results show that the proposed model consistently outperforms the baseline in terms of both Mean Squared Error (MSE) and Intersection over Union (IoU). In addition to the quantitative improvements, the qualitative results indicate more spatially concentrated and stable saliency maps, especially in scenes with multiple objects or cluttered backgrounds. This suggests that combining multi-scale processing with channel attention helps reduce the diffuse activations often observed in the baseline model.

From an application perspective, the proposed approach can be used in scenarios where estimating human visual attention is important, such as image compression, visual quality assessment, and general scene understanding. The improved localization of salient regions suggests that the model could be extended to more complex real-world settings, although further validation would be needed.

Despite these improvements, some limitations remain. The evaluation was performed only on the SALICON dataset, which limits conclusions about generalization to other domains with different data distributions. In addition, the model does not explicitly incorporate temporal information or higher-level contextual priors beyond spatial features, which could be relevant in more advanced saliency settings.

Throughout the project, attention was also given to practical aspects of model design and training. Working with encoder-decoder architectures highlighted how sensitive performance can be to small architectural changes, as well as the importance of balancing components such as multi-scale feature extraction and attention mechanisms. At the same time, data preprocessing and metric selection had a noticeable impact on the stability of the results.

Future work could explore more advanced loss functions, such as perceptual or adversarial objectives, which may better capture perceptual similarity in saliency prediction. Another interesting direction is the integration of transformer-based components to model long-range dependencies more explicitly, potentially complementing the convolutional backbone. Finally, evaluating the model across multiple datasets and real-world scenarios would provide a more complete assessment of robustness and generalization.

VIII. Takeaways and Challenges

A key outcome of this project was understanding how different architectural components can be combined within a deep learning framework to improve performance. In particular, the combination of multi-scale feature extraction and attention mechanisms proved effective when integrated into a U-Net baseline.

Another observation from the experiments is that increasing model complexity does not always lead to consistent improvements. Several design choices required iterative testing, and the effect of each component often depended on the others. This made understanding the architecture more important than simply tuning performance.

During development, several practical difficulties were encountered. The main challenge was integrating multiple modules into a stable encoder-decoder pipeline without causing training instability. This was further constrained by limited computational resources, as experiments were run on Google Colab with restricted time and memory.

Finally, the dataset split introduced an additional limitation. Since no dedicated test set was originally available, part of the validation set had to be used for final evaluation. This reduced the amount of data available for training and made performance estimates slightly less stable.

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